Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Consider any set of observations $x_1, x_2, x_3, \ldots, x_{101}$; it being given that $x_1 < x_2 < x_3 < \ldots < x_{100} < x_{101}$; then the mean
deviation of this set of observations about a point k is minimum when k equals

(A) x_1

(B) x_{51}

(C) $\frac{x_1+x_2+\ldots+x_{101}}{101}$

(D) x_{50}

Q.2

Let r be the range and $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$ be the S.D. of
set of observations $x_1, x_2, \ldots . x_n$, then

(A) $S \leq r\sqrt{\frac{n}{n-1}}$

(B) $S = r\sqrt{\frac{n}{n-1}}$

(C) $S \geq r\sqrt{\frac{n}{n-1}}$

(D) None of these

Q.3

For $(2n+1)$ observations $x_1, -x_1, x_2, -x_2, \ldots \ldots \ldots x_n, -x_n$ and
0 where x 's are all distinct. Let S.D. and M.D. denote the standard deviation and median respectively. Then which of the following is always true?

(A) S.D $<$ M.D.

(B) S.D. $>$ M.D.

(C) S.D.=M.D.

(D) Nothing can be said in general about the relationship of S.D. and M.D.

Q.4

Consider the first 10 positive integers. If we multiply each number by (-1) and then add 1 to each number, the variance of the numbers so obtained is

(A) 8.25

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(B) 6.5

(C) 387

(D) 287

Q.5

Coefficient of variation of two distributions are 50 and 60 and their arithmetic means are 30 and 25, respectively. Then, difference of their standard deviations is

(A) 0

(B) 1

(C) 1.5

(D) 2.5

Q.6

The mean and S.D. of the marks of 200 candidates were found to be 40 and 15 respectively. Later, it was discovered that a score of 40 was wrongly read as 50 . The correct mean and S.D. respectively are

(A) $14.98, 39.95$

(B) $39.95, 14.98$

(C) $39.95, 224.5$

(D) None of these

Q.7

The standard deviation of a variate x is σ. The standard deviation of the variable $\frac{ax+b}{c}$; a, b, c are constants, is

(A) $\left(\frac{a}{c}\right)\sigma$

(B) $\left|\frac{a}{c}\right|\sigma$

(C) $\left(\frac{a^2}{c^2}\right)\sigma$

(D) None

Q.8

The variance of 20 observations is 5 . If each observation is multiplied by 2, then the new variance of the resulting observation is

(A) $2^3 \times 5$

(B) $2^2 \times 5$

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(C) 2×5

(D) $2^4 \times 5$

Q.9

The mean deviation from the mean of the A.P.
$a, a+d, a+2d, \ldots \ldots \ldots a, a+2\text{nd}$ is

(A) $n(n+1)d$

(B) $\frac{n(n+1)d}{2n+1}$

(C) $\frac{n(n+1)d}{2n}$

(D) $\frac{n(n-1)d}{2n+1}$

Q.10

The mean and SD of 63 children on an arithmetic test are respectively 27.6 and 7.1. To them are added a new group of 26 who had less training and whose mean is 19.2 and $\text{SD is } 6.2$. The values of the combined group differ from the original as to (i) the mean and (ii) the SD is

(A) $25.1, 7.8$

(B) $2.3, 0.8$

(C) $1.5, 0.9$

(D) None of these

Q.11

The standard deviations of two sets containing 10 and 20 members are 2 and 3 respectively measured from their common mean 5 . The SD for the whole set of 30 members is

(A) $\frac{2}{\sqrt{3}}$

(B) $\sqrt{6}$

(C) $\sqrt{\left(\frac{22}{3}\right)}$

(D) $\sqrt{3}$

Q.12

The marks of some students were listed out of 75. The SD of marks was found to be 9. Subsequently the marks were raised to a maximum of 100 and variance of new marks was calculated. The new variance is

(A) 144

(B) 122

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(C) 81

(D) None of these

Q.13

If the variable takes values $0, 1, 2, 3, \ldots, n$ with frequencies proportional to ${}^nC_0, {}^nC_1, {}^nC_2, \ldots, {}^nC_n$ respectively, the variance is

(A) $\frac{n}{4}$

(B) $\frac{n}{3}$

(C) $\frac{2n}{5}$

(D) None of these

Q.14

The standard deviation of n observations $x_1, x_2, \ldots, x_n$ is 2. If $\sum_{i=1}^{n} x_i = 20$ and $\sum_{i=1}^{n} x_i^2 = 100$, then n is

(A) 10 or 20

(B) 5 or 10

(C) 5 or 20

(D) 5 or 15

Q.15

All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given ?

(A) mean

(B) median

(C) mode

(D) variance

Answers & Solutions

Q.1 Answer:

x_{51}

Solution:

Mean deviation is minimum when it is considered about the item, equidistant from the beginning and the end i.e.

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the median. In this case median is $\frac{101+1}{2}$ th i.e. 51st item i.e., x_{51}.

Q.2 Answer:

$S \leq r\sqrt{\frac{n}{n-1}}$

Solution:

We have $r = \max_{i \neq j} |x_i - x_j|$ **and** $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

Now, consider $(x_i - \bar{x})^2 = \left(x_i - \frac{x_1+x_2+\ldots..+x_n}{n}\right)^2$

$= \frac{1}{n^2}[(x_i - x_1) + (x_i - x_2) + \ldots. + (x_i - x_i - 1)]$

$+ (x_i - x_i + 1) + \ldots + (x_i - x_n)] \leq \frac{1}{n^2}[(n-1)r]^2$

$[\because |x_i - x_j| \leq r]$

$\Rightarrow (x_i - \bar{x})^2 \leq r^2 \Rightarrow \sum_{i=1}^{n}(x_i - \bar{x})^2 \leq nr^2$

$\Rightarrow \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \leq \frac{nr^2}{(n-1)} \Rightarrow S^2 \leq \frac{nr^2}{(n-1)}$

$\Rightarrow S \leq r\sqrt{\frac{n}{n-1}}.$

Q.3 Answer:

S.D. $>$ M.D.

Solution:

On arranging the given observations in ascending order, we get All negative terms $\underbrace{0}_{(n+1)^{th}\text{ term}}$ **All positive terms The median of given observations** $= (n+1)^{\text{th}}$ **term** $= 0$ **S. S.D** $>$ M.D

Q.4 Answer:

8.25

Solution:

First ten positive integers are $1, 2, 3, 4, 5, 6, 7, 8, 9$ **and** 10 **.**

Sum of these numbers $\left(\sum x_i\right) = 1 + 2 + \ldots + 10 = 55$

Sum of squares of these numbers $\left(\sum x_i^2\right) = 1^2 + 2^2 + \ldots + 10^2 = 385$

Standard deviation $(\sigma) = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{1}{n}\sum x_i\right)^2}$

$= \sqrt{\frac{385}{10} - (5.5)^2} = \sqrt{38.5 - 30.25} = \sqrt{8.25}$

$\therefore$ **Variance** $\left(\sigma^2\right) = 8.25$

Q.5 Answer:

0

Solution:

We know that,

Coefficient of variation $= \frac{\sigma}{\bar{x}} \times 100$

$\therefore$ CV **of 1 st distribution** $= \frac{\sigma_1}{30} \times 100$

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$\Rightarrow \quad 50 = \frac{\sigma_1}{30} \times 100 \quad [\text{CV of 1 st distribution} = 50 \text{ (given)}]$

$\Rightarrow \quad \sigma_1 = 15$

Also, CV of 2 nd distribution $= \frac{\sigma_2}{25} \times 100$

$\Rightarrow 60 = \frac{\sigma_2}{25} \times 100 \Rightarrow \sigma_2 = \frac{60 \times 25}{100} \Rightarrow \sigma_2 = 15$

Thus, $\sigma_1 - \sigma_2 = 15 - 15 = 0$

Q.6 Answer:

$39.95, 14.98$

Solution:

Corrected $\Sigma x = 40 \times 200 - 50 + 40 = 7990$

$\therefore$ **Corrected** $\bar{x} = 7990/200 = 39.95$

Incorrect $\Sigma x^2 = n\left[\sigma^2 + \bar{x}^2\right] = 200\left[15^2 + 40^2\right]$

$= 365000$

Corrected $\Sigma x^2 = 365000 - 2500 + 1600 = 364100$

$\therefore$ **Corrected** $\sigma = \sqrt{\frac{364100}{200} - (39.95)^2}$

$= \sqrt{(1820.5 - 1596)} = \sqrt{224.5} = 14.98.$

Q.7 Answer:

$\left|\frac{a}{c}\right| \sigma$

Solution:

Let $\mathrm{y} = \frac{\mathrm{ax}+\mathrm{b}}{\mathrm{c}} \Rightarrow \mathrm{y} = \frac{\mathrm{a}}{\mathrm{c}}\mathrm{x} + \frac{\mathrm{b}}{\mathrm{c}}$

$\Rightarrow y = Ax + B$ **where** $A = \frac{a}{c}$ **and** $B = \frac{b}{c}$

So, $\bar{y} = A\bar{x} + B$ **and hence**

$y - \bar{y} = Ax + B - (A\bar{x} + B) = A(x - \bar{x})$

$\Rightarrow (\mathrm{y} - \overline{\mathrm{y}})^2 = \mathrm{A}^2(\mathrm{x} - \overline{\mathrm{x}})^2$

$\Rightarrow \sum(y - \bar{y})^2 = A^2 \sum(x - \bar{x})^2$

$\Rightarrow \mathrm{n}\sigma_{\mathrm{y}}^2 = \mathrm{A}^2\left(\mathrm{n}\sigma_{\mathrm{x}}^2\right) \Rightarrow \sigma_{\mathrm{y}} = |\mathrm{A}|\sigma_{\mathrm{x}}$

Hence standard deviation is multiplied by $|\mathrm{A}| = \left|\frac{\mathrm{a}}{\mathrm{c}}\right|$

Q.8 Answer:

$2^2 \times 5$

Solution:

Let the observations be $x_1, x_2, \ldots, x_{20}$ **and** $\bar{x}$ **be their mean. Given that, variance** $= 5$ **and** $\mathrm{n} = 20$**. We know that,**

Variance $\left(\sigma^2\right) = \frac{1}{\mathrm{n}} \sum_{\mathrm{i}=1}^{20} \left(\mathrm{x_i} - \overline{\mathrm{x}}\right)^2$

i.e. $5 = \frac{1}{20} \sum_{\mathrm{i}=1}^{20} \left(\mathrm{x_i} - \overline{\mathrm{x}}\right)^2$ **or** $\sum_{\mathrm{i}=1}^{20} \left(\mathrm{x_i} - \overline{\mathrm{x}}\right)^2 = 100 \quad \ldots$ **(i)**

If each observation is multiplied by 2 and the new resulting observations are $\mathrm{y_i}$**, then**

$\mathrm{y_i} = 2\mathrm{x_i}$ **i.e.,** $\mathrm{x_i} = \frac{1}{2}\mathrm{y_i}$

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Therefore, $\overline{\mathrm{y}}=\frac{1}{\mathrm{n}} \sum_{\mathrm{i}=1}^{20} \mathrm{y}_{\mathrm{i}}=\frac{1}{20} \sum_{\mathrm{i}=1}^{20} 2 \mathrm{x}_{\mathrm{i}}=2 \cdot \frac{1}{20} \sum_{\mathrm{i}=1}^{20} \mathrm{x}_{\mathrm{i}}$

i.e., $\overline{\mathrm{y}}=2 \overline{\mathrm{x}}$ **or** $\overline{\mathrm{x}}=\frac{1}{2} \overline{\mathrm{y}}$

On substituting the values of x_{i} **and** $\overline{\mathrm{x}}$ **in eq. (i), we get**

$\sum_{i=1}^{20}\left(\frac{1}{2} y_i-\frac{1}{2} \bar{y}\right)^2=100$ **i.e.,** $\sum_{i=1}^{20}\left(y_i-\bar{y}\right)^2=400$

Thus, the variance of new observations

$=\frac{1}{20} \times 400=20=2^2 \times 5$

Q.9 Answer:

$\frac{n(n+1)d}{2n+1}$

Solution:

The mean of the series

$$\overline{\mathrm{X}}=\frac{1}{2n+1}\{a+(a+d)+(a+2d)+\ldots..+(a+2nd)\}$$

$$=\frac{1}{2n+1}\left\{\frac{2n+1}{2}(2a+2nd)\right\}=a+nd$$

Therefore, mean deviation from mean

$\frac{1}{2n+1} \sum_{r=0}^{2n}|(a+rd)-(a+nd)|=\frac{1}{2n+1} \sum_{r=0}^{2n}|r-n| d$

$=\frac{[2(1+2+\ldots+n)+0] d}{2n+1}=\frac{n(n+1)d}{2n+1}$

Q.10 Answer:

$25.1, 7.8$

Solution:

Mean and SD σ **of the combined group are**

$m=\frac{63 \times 27.6+26 \times 19.2}{63+26}=25.1$

Thus, AM is decreased by $27.6-25.1=2.5$

$\sigma^2=\frac{63 \times(7.1)^2+26 \times(6.2)^2}{89}$

$+\frac{63(25.1-27.6)^2+26(25.1-19.2)^2}{89}$

$\Rightarrow \sigma=7.8($ **approx** $)$

Hence, (a) is the correct answer.

Q.11 Answer:

$\sqrt{\left(\frac{22}{3}\right)}$

Solution:

Using, $\sigma=\sqrt{\frac{n_1\left(\sigma_1^2+d_1^2\right) n_2\left(\sigma_2^2+d_2^2\right)}{n_1+n_2}}=\sqrt{\frac{22}{3}}$

Q.12 Answer:

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Solution:

Given, $\sigma = 9$ Let a student obtains x out of 75 . Then his marks out of 100 are $\frac{4x}{3}$. Each observation is multiplied by $\frac{4}{3}$
New $\sigma = \frac{4}{3} \times 9 = 12$, Variance $= \sigma^2 = 144$

Q.13 Answer:

$\frac{n}{4}$

Solution:

$\mu_1' = \frac{\sum_{r=0}^{n} r.{}^nC_r}{\sum_{r=0}^{n} {}^nC_r} = \frac{\sum_{r=0}^{n} r.\frac{n}{r}n^{n-1}C_{r-1}}{\sum_{r=0}^{n} {}^nC_r}$
$= \frac{n\sum_{r=1}^{n} {}^{n-1}C_{r-1}}{\sum_{r=0}^{n} {}^nC_r} = \frac{n.2^{n-1}}{2^n} = \frac{n}{2}$
$\mu_2' = \frac{\sum_{r=0}^{n} r^2.{}^nC_r}{\sum_{r=0}^{n} {}^nC_r} = \frac{1}{2^n}\sum_{r=0}^{n}\{r(r-1)+r\}^nC_r$
$= \frac{1}{2^n}\sum_{r=0}^{n} r(r-1)^nC_r + \frac{n}{2}$
$= \frac{1}{2^n}\sum_{r=0}^{n} r(r-1)\frac{n(n-1)}{r(r-1)}n-2C_{r-2} + \frac{n}{2}$
$= \frac{n(n-1)}{2^n}\cdot 2^{n-2} + \frac{n}{2}$
$= \frac{n(n-1)}{4} + \frac{n}{2}$
Variance $\mu_2 = \mu_2' - (\mu_1')^2$
$= \frac{n(n-1)}{4} + \frac{n}{2} - \left(\frac{n}{2}\right)^2 = \frac{n}{4}$
Hence (a) is the correct answer

Q.14 Answer:

5 or 20

Solution:

We know $\mathrm{SD} = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2}$
$\therefore 2 = \sqrt{\frac{100}{n} - \left(\frac{20}{n}\right)^2} = \sqrt{\frac{100}{n} - \frac{400}{n^2}}$
$\Rightarrow 4 = \frac{100}{n} - \frac{400}{n^2}$
$\Rightarrow n^2 - 25n + 100 = 0 \Rightarrow n = 20, 5$

Q.15 Answer:

variance

Solution:

If initially all marks were x_i then $\sigma_\mathrm{i}^2 = \frac{\sum(x_i-\bar{x})^2}{N}$
Now each is increased by 10
$\sigma_i^2 = \frac{\sum[(x_i+10)-(\bar{x}+10)]^2}{N} = \frac{\sum_i(x_i-\bar{x})^2}{N} = \sigma_i^2$
Hence, variance will not change even after the grace marks were given.

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